

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Distributed Systems Aug 2025 to May 2026

- Reduced rebuffering by **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configurations.
- Cut experiment setup effort by about **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making large experiment sweeps reproducible.
- Built and exercised **14 interoperability scenarios** covering networking, dependency, and **failure recovery** behavior in a distributed real-time system.
- Built and evaluated **RAG** pipelines (chunking, embeddings, vector retrieval, top-k, grounded prompts), separating **retrieval** failures from generation failures to guide fixes.

Turito | Software Engineering Intern, Backend Reliability Jun 2022 to Oct 2022

- Cut controlled-disruption failures by **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Slashed p95 **CPU** by **22%** by profiling high-frequency media paths and removing excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Reduced production diagnosis time by **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Lowered network egress by **14%** by removing redundant media-path processing and simplifying routing and payload handling.
- Worked across a **WebRTC** control plane and **RTP/RTCP** media paths spanning about **12 regions**, including ICE/STUN/TURN connection setup and degraded-network handling.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Surfaced backend fixes that raised equivalent-load signaling throughput by about **31%**.

TECHNICAL SKILLS

Languages: Python, Java, C++, Rust, TypeScript, SQL

Backend & Distributed: FastAPI, Flask, REST, gRPC, Kafka, Temporal, WebSockets, asyncio, Tokio

Data & Cloud: PostgreSQL, Redis, SQLite, MongoDB, Docker, AWS, Linux, CI/CD, pytest

Frontend & AI: React, TypeScript, Next.js, PyTorch, RAG, OpenAI and Anthropic APIs

PROJECTS

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Built an OpenAI-compatible gateway with exact and **semantic caching**, provider **failover**, and rate limiting that cut p50 latency from **759 ms to 44 ms** over a 4,000-request load test.

Portcullis, Policy Gateway | *Rust*

- Built a **Rust** proxy that centralizes authorization and rate limiting before requests reach services and emits auditable accept, reject, and throttle decisions.

Proxima, Vector Search Engine | *C++17, AVX2/SIMD, HNSW, pybind11*

- Built an HNSW vector search engine from scratch that reaches **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** faster than hnswlib and **2.5x** faster than FAISS on 100K vectors.

OPEN SOURCE

- corsairdev/corsair**: 6 merged **webhook signature verification** fixes (Telegram, Teams, SharePoint, Monday, GitLab, Jira) that fail closed
- google/osv-scalibr**: upstream change to Google's software composition analysis library (merged)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Distributed Computing, Advanced Algorithms, Database Systems, Computer Networks, Machine Learning, NoSQL

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)